

NOXUS — WEEK 5 & WEEK 6

WEEK 5: DOMAIN SPECIALIZATION & STRUCTURED LEARNING

1. WEEK 5 OVERVIEW

Week Name: Domain Specialization & Structured Learning

Duration: 5–7 Days

Main Objective

After exploring multiple domains in Week 3 and working in teams during Week 4, students now begin focusing on their selected technical domain.

Each student will select a **Primary Domain** and optionally a **Secondary Domain**.

The goal is not to make students experts in one week. The goal is to provide a structured starting path and help them build stronger practical skills.

2. WEEK 5 GOALS

By the end of Week 5, students should:

- Select or confirm their primary technical domain.
 - Follow a structured domain-specific learning path.
 - Complete domain-based training modules.
 - Learn essential tools used in that domain.
 - Build a small individual technical activity.
 - Continue coding and problem-solving practice where applicable.
 - Document their learning.
 - Update their GitHub portfolio.
 - Earn XP based on learning, consistency, and practical work.
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3. AVAILABLE DOMAIN TRACKS

Students can select one primary track.

TRACK 1 — WEB DEVELOPMENT

Topics

- HTML and CSS fundamentals.
- JavaScript basics.
- Responsive design.
- Components.
- Basic frontend project structure.
- Introduction to APIs.
- Introduction to frontend frameworks.

Practical Activity

Build a small responsive website or web application.

TRACK 2 — APP DEVELOPMENT

Topics

- Mobile application basics.
- User interface.
- Navigation.
- State and data handling.
- APIs.
- Application structure.

Practical Activity

Create a basic mobile application prototype or simple functional application.

TRACK 3 — AI / MACHINE LEARNING

Topics

- Introduction to AI and ML.
- Datasets.
- Features and labels.
- Data preprocessing.
- Training and testing.
- Basic models.
- Model evaluation.

Practical Activity

Train and test a simple machine learning model.

TRACK 4 — DATA SCIENCE

Topics

- Introduction to datasets.
- Data cleaning.
- Data analysis.
- Visualization.
- Basic statistics.
- Finding insights.

Practical Activity

Analyze a small dataset and create meaningful visualizations.

TRACK 5 — EMBEDDED SYSTEMS / IOT

Topics

- Microcontroller basics.
- Sensors and actuators.
- GPIO.
- Communication basics.
- IoT architecture.
- Cloud or dashboard introduction.

Practical Activity

Build or simulate a simple sensor-based system.

TRACK 6 — ROBOTICS / AUTOMATION

Topics

- Sensors.
- Actuators.
- Motor control.
- Control logic.
- Automation systems.
- Basic robotics architecture.

Practical Activity

Create a basic robotic or automation simulation/prototype.

TRACK 7 — UI/UX DESIGN

Topics

- Design thinking.
- User research basics.
- Wireframes.
- User flows.
- Components.
- Basic prototyping.

Practical Activity

Create a complete user flow and prototype for a simple application.

TRACK 8 — CYBERSECURITY

Topics

- Cybersecurity fundamentals.
- Networking basics.
- Web security awareness.
- Authentication concepts.
- Common vulnerabilities.
- Safe security practices.

Practical Activity

Complete a safe beginner-level security analysis or capture-the-flag style learning exercise.

TRACK 9 — CAD / MECHANICAL / MECHATRONICS

Topics

- Design fundamentals.
- CAD modeling.
- Components and assemblies.
- Mechanisms.
- Basic manufacturing considerations.
- Automation concepts.

Practical Activity

Design a simple component, assembly, mechanism, or automation concept.

4. DAY-WISE PLAN

DAY 1 — DOMAIN SELECTION & ROADMAP

Training Topic

Choosing Your Learning Path

Students will:

- Review their Week 3 exploration results.
- Review their interests.
- Review mentor recommendations.
- Select a primary domain.
- Select an optional secondary domain.

Student Activity

Each student creates a simple learning plan.

Example:

```
Current Level
  ↓
Week 5 Learning Goals
  ↓
Skills to Learn
  ↓
Tools to Explore
  ↓
Practical Activity
  ↓
Portfolio Update
```

Outcome

Every student begins a structured domain-specific learning journey.

DAY 2 — CORE DOMAIN TRAINING

Students attend domain-specific training sessions.

Each track should contain:

- Core concepts.
- Required tools.
- Basic workflow.

- Beginner practical examples.

Students should complete at least one guided exercise.

Tasks

- [] Complete training module.
 - [] Set up required tools.
 - [] Complete guided exercise.
 - [] Document learning.
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DAY 3 — HANDS-ON DOMAIN PRACTICE

Students begin working independently.

Activity Requirements

Each student must:

1. Complete a practical task.
2. Use at least one domain-specific tool.
3. Document the implementation.
4. Record challenges faced.
5. Push code or project files to GitHub where applicable.

Outcome

Students move from guided learning toward independent exploration.

DAY 4 — PROBLEM-BASED DOMAIN CHALLENGE

Students receive a small domain-specific challenge.

Example:

Web Development

Create a responsive interface for a student-focused problem.

AI/ML

Build a basic classification or prediction model.

IoT

Design a sensor-based monitoring solution.

UI/UX

Create a prototype for a real user problem.

CAD

Design a simple functional component.

Data Science

Analyze a dataset and identify useful insights.

Students should focus on:

- Understanding the problem.
 - Selecting appropriate tools.
 - Building a basic solution.
 - Documenting the approach.
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DAY 5 — REVIEW & PORTFOLIO UPDATE

Students complete:

- Final testing.
- Documentation.
- README.
- GitHub updates.
- Screenshots or demo.
- Learning reflection.

Reflection Questions

- What did I learn?
 - What was difficult?
 - How did I solve the problem?
 - What do I want to learn next?
 - What is my confidence level now?
-

5. WEEK 5 CODING PRACTICE

Students continue problem-solving according to their selected programming track.

Suggested Topics

- Arrays.
- Strings.
- Hash maps/dictionaries.

- Searching.
- Sorting basics.
- Recursion basics.

Practice Structure

- 2–3 easy problems.
- Topic-based practice.
- One challenge problem.
- Optional advanced problem.

Students continue syncing accepted solutions to GitHub and maintaining their solution explanations.

6. WEEK 5 XP SYSTEM

Activity	Suggested XP
Confirm domain	10 XP
Complete training module	25 XP
Guided exercise	30 XP
Domain practical	60 XP
Domain challenge	75 XP
Documentation	15 XP
Portfolio update	20 XP
Coding problem	10–25 XP
Complete Week 5	100 XP

7. WEEK 5 COMPLETION CRITERIA

Students must:

- ☐ Select or confirm primary domain.
 - ☐ Complete core domain training.
 - ☐ Set up required tools.
 - ☐ Complete a guided exercise.
 - ☐ Complete an independent practical.
 - ☐ Complete the domain challenge.
 - ☐ Submit documentation.
 - ☐ Update GitHub or portfolio.
 - ☐ Continue coding practice where applicable.
 - ☐ Complete Week 5 reflection.
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8. WEEK 5 FINAL OUTCOME

By the end of Week 5, every student should have:

1. A clearly defined primary technical domain.
 2. Experience using domain-specific tools.
 3. Completed guided and independent practical activities.
 4. A documented technical challenge.
 5. Updated GitHub or portfolio work.
 6. Improved understanding of their selected specialization.
 7. A clear direction for deeper learning.
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WEEK 6: REAL-WORLD DEVELOPMENT & ADVANCED COLLABORATION

1. WEEK 6 OVERVIEW

Week Name: Real-World Development & Advanced Collaboration

Duration: 5–7 Days

Main Objective

Week 6 focuses on moving students from small guided activities toward a more realistic development environment.

Students will work in teams or domain groups to solve a practical problem while learning professional workflows.

2. WEEK 6 GOALS

By the end of Week 6, students should:

- Understand real-world project workflows.
 - Learn requirement gathering.
 - Define project scope.
 - Create a basic MVP.
 - Use task management.
 - Work collaboratively using GitHub.
 - Understand code and work reviews.
 - Handle feedback and iteration.
 - Build a functional prototype.
 - Present progress professionally.
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3. DAY 1 — REAL-WORLD PROJECT WORKFLOW

Training Topic

How Professional Projects Are Planned

Topics Covered

- Understanding requirements.
- Identifying stakeholders.
- Defining users.
- Scope management.
- MVP definition.
- Feature prioritization.
- Deadlines.
- Task management.

Project Planning Flow

```
graph TD; Problem --> Research; Research --> Requirements; Requirements --> Users; Users --> Features; Features --> MVP; MVP --> Task_Breakdown[Task Breakdown]; Task_Breakdown --> Development;
```

Problem
↓
Research
↓
Requirements
↓
Users
↓
Features
↓
MVP
↓
Task Breakdown
↓
Development

Student Activity

Teams convert a problem statement into a project plan.

DAY 2 — TASK MANAGEMENT & PROJECT ORGANIZATION

Training Topic

Planning Before Building

Students learn:

- Breaking features into tasks.
- Creating issues.
- Assigning responsibilities.
- Setting priorities.
- Tracking progress.
- Managing blockers.
- Writing clear task descriptions.

Suggested Workflow

Backlog

↓

To Do

↓

In Progress

↓

Review

↓

Testing

↓

Completed

Activity

Each team creates a project board and assigns tasks to members.

DAY 3 — DEVELOPMENT SPRINT

Teams begin building their MVP.

Each student should have:

- Clearly assigned tasks.
- Defined deliverables.
- Repository access where applicable.
- A documented contribution.

Mentors can monitor:

- Progress.
 - Blockers.
 - Team communication.
 - Technical decisions.
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DAY 4 — REVIEW, TESTING & ITERATION

Teams review the work completed.

Activities

- Code or design review.
- Functional testing.
- Bug identification.
- Feedback collection.
- Improvement planning.
- Task reassignment where required.

Students learn that professional projects are iterative.

Development Cycle

Build

↓

Review

↓

Test

↓

Find Issues

↓

Improve

↓

Repeat

DAY 5 — SPRINT DEMO

Each team demonstrates its current MVP.

Demo Structure

1. Problem being solved.
2. Target users.
3. MVP features.

4. Technology used.
 5. Team contributions.
 6. Live demonstration.
 7. Challenges.
 8. Feedback received.
 9. Next improvements.
-

4. WEEK 6 TEAM PROJECT RULES

Every project should have:

- A defined problem.
 - Clear target users.
 - A realistic MVP.
 - A project repository or workspace.
 - Task assignments.
 - Documentation.
 - Contribution tracking.
 - Testing.
 - A final demo.
-

5. CONTRIBUTION EVALUATION

Student contribution should be calculated using multiple factors.

Contribution Score

Task Completion

+

Work Quality

+

Documentation

+

Collaboration

+

Review Participation

+

Mentor Feedback

GitHub commits should be considered as evidence of activity but should not be the only evaluation method.

6. WEEK 6 XP SYSTEM

Activity	Suggested XP
Project planning	30 XP
Create/complete assigned task	25–50 XP
Review contribution	20 XP
Testing contribution	25 XP
Documentation	20 XP
MVP contribution	75 XP
Sprint demo	50 XP
Team completion bonus	50 XP
Complete Week 6	125 XP

7. WEEK 6 STUDENT DASHBOARD

WEEK 6 – REAL-WORLD DEVELOPMENT SPRINT

Project: Smart Campus Solution

Role: Frontend Developer

Assigned Tasks: 5

Completed: 4

Tasks In Review: 1

Personal Contribution: 82%

Team MVP Progress: 76%

Week XP: 410 XP

Current Status: On Track

8. WEEK 6 COMPLETION CHECKLIST

Every student must:

- ☐ Participate in requirement analysis.
- ☐ Help define the MVP.
- ☐ Accept assigned tasks.
- ☐ Complete assigned responsibilities.
- ☐ Participate in reviews.
- ☐ Participate in testing.
- ☐ Update documentation.
- ☐ Attend sprint review.
- ☐ Participate in the project demo.

9. WEEK 6 FINAL OUTCOME

By the end of Week 6, students should have:

1. Experience working on a realistic project workflow.
2. Experience converting requirements into tasks.
3. Experience defining an MVP.
4. Experience using collaborative project management.
5. A functional prototype or MVP contribution.
6. Experience receiving and applying feedback.
7. A measurable contribution record.
8. Stronger teamwork and communication skills.

WEEK 5 & 6 JOURNEY

WEEK 5

Domain Selection



Core Training



Guided Practice

↓

Independent Practice

↓

Domain Challenge

↓

Portfolio Update

WEEK 6

Problem Selection

↓

Requirement Analysis

↓

MVP Planning

↓

Task Breakdown

↓

Development Sprint

↓

Review & Testing

↓

Iteration

↓

Sprint Demo

↓

READY FOR THE NEXT PHASE: PROJECT BUILDING, ADVANCED SKILLS & PROFESSIONAL DEVELOPMENT